

Distance Formula & Simplifying Radicals — Answer Key

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Date: May 15, 2026

Each problem shows the Answer (green) and a short Solution.

Self-check rule. Try the worksheet first. Only after you've written your answer should you flip here. For each one you missed, identify *where* the error happened — arithmetic, sign, or the wrong rule applied.

Part A — Distance Formula

1. $(1, 2)$ and $(6, 3)$

ANSWER $\sqrt{26}$

SOLUTION $d = \sqrt{(6-1)^2 + (3-2)^2} = \sqrt{25+1} = \sqrt{26}$.

2. $(0, -6)$ and $(7, 1)$

ANSWER $7\sqrt{2}$

SOLUTION $d = \sqrt{(7-0)^2 + (1-(-6))^2} = \sqrt{49+49} = \sqrt{98} = 7\sqrt{2}$.

3. $(-3, 4)$ and $(5, -2)$

ANSWER 10

SOLUTION $d = \sqrt{(5-(-3))^2 + (-2-4)^2} = \sqrt{64+36} = \sqrt{100} = 10$. (Classic 6-8-10 right triangle.)

4. $(-5, 7)$ and $(-7, 8)$

ANSWER $\sqrt{5}$

SOLUTION $d = \sqrt{(-7-(-5))^2 + (8-7)^2} = \sqrt{4+1} = \sqrt{5}$.

5. $(0, 0)$ and $(3, 4)$

ANSWER 5

SOLUTION $d = \sqrt{3^2 + 4^2} = \sqrt{9+16} = \sqrt{25} = 5$. (The classic 3-4-5 right triangle.)

6. $(2, -1)$ and $(2, 6)$

ANSWER 7

SOLUTION Same x — purely vertical: $d = |6 - (-1)| = 7$.

7. $(-4, 3)$ and $(8, 3)$

ANSWER 12

SOLUTION Same y — purely horizontal: $d = |8 - (-4)| = 12$.

8. Perimeter of triangle with vertices $(0, 0)$, $(3, 0)$, $(0, 4)$.

ANSWER 12

SOLUTION Sides: 3 (horizontal), 4 (vertical), $\sqrt{9 + 16} = 5$ (hypotenuse). Perimeter = $3 + 4 + 5 = 12$.

9. $(2, 5)$ and $(x, 5)$ are 6 units apart. Find x .

ANSWER $x = -4$ or $x = 8$

SOLUTION Same $y \rightarrow |x - 2| = 6 \Rightarrow x - 2 = \pm 6 \Rightarrow x = 8$ or $x = -4$.

10. $A(-1, 2)$, $B(4, 2)$, $C(4, 6)$ — find AC .

ANSWER $\sqrt{41}$

SOLUTION $AC = \sqrt{(4 - (-1))^2 + (6 - 2)^2} = \sqrt{25 + 16} = \sqrt{41}$. (Hypotenuse of right triangle with legs 5 and 4.)

Part B — Simplifying Radicals

1. $\sqrt{196x^6}$

ANSWER $14x^3$

SOLUTION $\sqrt{196} = 14$ (since $14^2 = 196$); $\sqrt{x^6} = x^3$.

2. $\sqrt{24x^9}$

ANSWER $2x^4\sqrt{6x}$

SOLUTION $24 = 4 \cdot 6$, $x^9 = x^8 \cdot x$. So $\sqrt{4 \cdot 6 \cdot x^8 \cdot x} = 2x^4\sqrt{6x}$.

3. $\sqrt{180x^{36}y^{25}}$

ANSWER $6x^{18}y^{12}\sqrt{5y}$

SOLUTION $180 = 36 \cdot 5$; x^{36} has even exponent $\rightarrow$ fully out; $y^{25} = y^{24} \cdot y$. So $\sqrt{36 \cdot 5 \cdot x^{36} \cdot y^{24} \cdot y} = 6x^{18}y^{12}\sqrt{5y}$.

4. $2x\sqrt{12x^{35}y^{16}}$

ANSWER $4x^{18}y^8\sqrt{3x}$

SOLUTION $12 = 4 \cdot 3$; $x^{35} = x^{34} \cdot x$; y^{16} even. So $\sqrt{12x^{35}y^{16}} = 2x^{17}y^8\sqrt{3x}$. Multiply by $2x$: $2x \cdot 2x^{17}y^8\sqrt{3x} = 4x^{18}y^8\sqrt{3x}$.

5. $\sqrt{50}$

ANSWER $5\sqrt{2}$

SOLUTION $50 = 25 \cdot 2 \Rightarrow \sqrt{50} = 5\sqrt{2}$.

6. $\sqrt{72x^4}$

ANSWER $6x^2\sqrt{2}$

SOLUTION $72 = 36 \cdot 2$; $\sqrt{x^4} = x^2$. So $\sqrt{72x^4} = 6x^2\sqrt{2}$.

7. $\sqrt{8x^3} \cdot 3\sqrt{12x}$

ANSWER $12x^2\sqrt{6}$

SOLUTION Combine inside one radical: $3\sqrt{8x^3 \cdot 12x} = 3\sqrt{96x^4}$. $96 = 16 \cdot 6$, $\sqrt{x^4} = x^2$. So $= 3 \cdot 4x^2\sqrt{6} = 12x^2\sqrt{6}$.

8. $\sqrt{45x^5y^3}$

ANSWER $3x^2y\sqrt{5xy}$

SOLUTION $45 = 9 \cdot 5$; $x^5 = x^4 \cdot x$; $y^3 = y^2 \cdot y$. So $\sqrt{9 \cdot 5 \cdot x^4 \cdot x \cdot y^2 \cdot y} = 3x^2y\sqrt{5xy}$.

9. $\sqrt{98x^7y^{10}}$

ANSWER $7x^3y^5\sqrt{2x}$

SOLUTION $98 = 49 \cdot 2$; $x^7 = x^6 \cdot x$; y^{10} even. So $\sqrt{49 \cdot 2 \cdot x^6 \cdot x \cdot y^{10}} = 7x^3y^5\sqrt{2x}$.

10. $\sqrt{8x^3}(3\sqrt{12x} - 4x\sqrt{18x^9})$

ANSWER $12x^2\sqrt{6} - 48x^7$

SOLUTION Distribute first.

- First piece: $\sqrt{8x^3} \cdot 3\sqrt{12x} = 3\sqrt{96x^4} = 3 \cdot 4x^2\sqrt{6} = 12x^2\sqrt{6}$.
- Second piece: $\sqrt{8x^3} \cdot 4x\sqrt{18x^9} = 4x\sqrt{8 \cdot 18 \cdot x^{12}} = 4x\sqrt{144x^{12}} = 4x \cdot 12x^6 = 48x^7$.
- Combine: $12x^2\sqrt{6} - 48x^7$.